Statistical Efficiency of Birth-Death Tree Parameter Estimation

Stat 700

Santosh Desai, Matt McAnear, Urvi Mehta

Department of Statistics, University of Michigan

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The Phylodeep Package

Phylodeep Paper is byt J. Voznica et al [1]

- Parameter estimation is difficult and MLE method can be challenging for larger trees.
- Major components of the paper to address this
  - CBLV representation of trees
  - Pre-trained neural networks for pre-specified tree types
- The goal is to speed up inference

Usage

- Input: Phylogenetic tree in Newick format
- Output: Estimated parameters (e.g., birth rate (λ), death rate (μ), R_0)
- Given a tree, Phylodeep selects the pre-trained model with the highest probability
- Outputs actually come with uncertainty estimates from a parametric bootstrap

Maximum Likelihood Estimation

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Phylodeep Methodology

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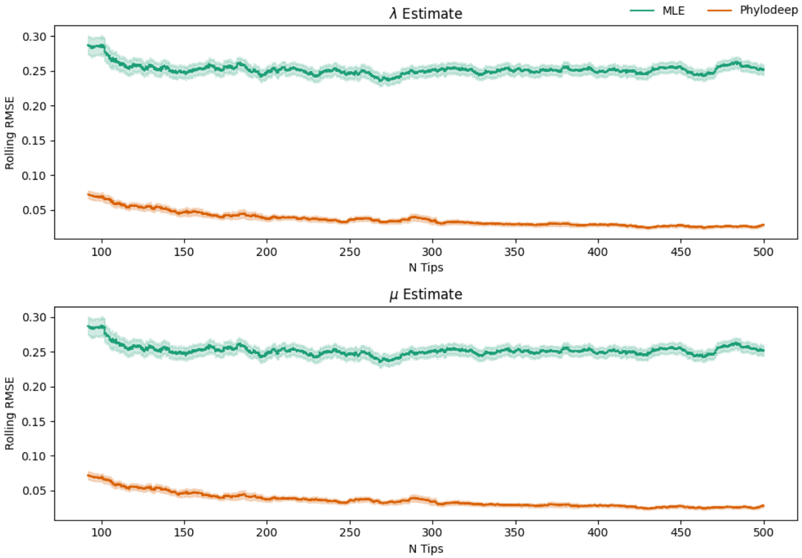

Figure: Rolling bootstrapped RMSE estimates of μ and λ by tree size and resulting 95% CI. Window size is 500.

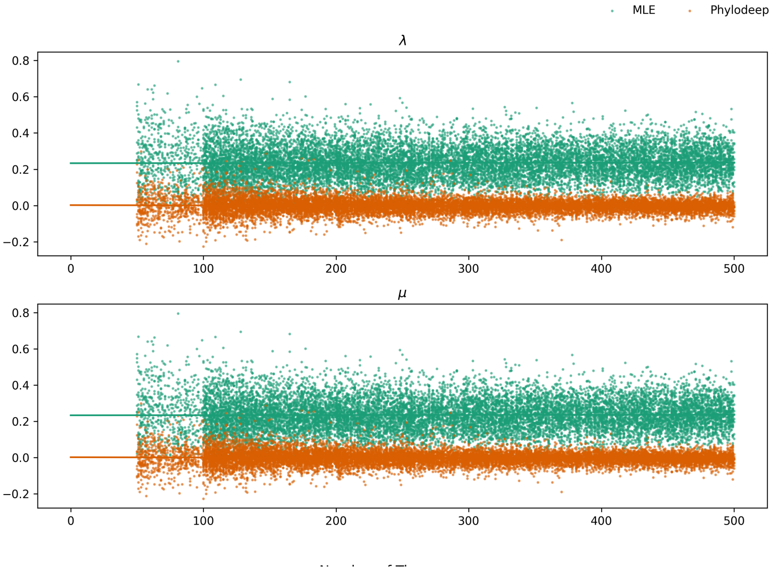

Figure: Raw plot of the absolute errors in λ, μ by tree size.

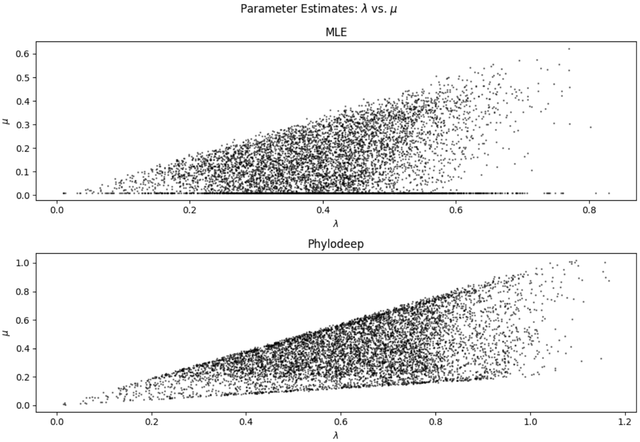

Figure: Plot of λ vs μ estimates to show distribution of estimates.

References

[1] J. Voznica et al. “Deep learning from phylogenies to uncover the epidemiological dynamics of outbreaks”. en. In: *Nature Communications* 13.1 (July 2022). Publisher: Nature Publishing Group, p. 3896. ISSN: 2041-1723. DOI: 10.1038/s41467-022-31511-0. URL: https://www.nature.com/articles/s41467-022-31511-0 (visited on 11/04/2025).